

Jointly Distributed Random Variables:

So far, we have touched on settings where there are multiple r.v.'s of interest.

Ex: $X \sim \text{Bin}(2, 1/2)$ consists of two "independent" $\text{Ber}(1/2)$ random variables.

Now, we will (finally) seek to answer:

i) How can we describe the distribution of two (or more) random variables?

ii) What does it mean for two (or more) r.v.'s to be independent?

iii) How can we measure "dependence" b/w two r.v.'s?

Q: Why might we care about this?

A: Several possible reasons

1- Maybe we observe, for one object, two or more different quantities. e.g. the height and weight, and want to know how this pair of quantities can vary.

2- More importantly for us, we think of each r.v. as corresponding to some variable of interest for a single object. E.g. the tensile strength of a particular beam is a $N(200, 10)$ r.v., in MPa. Having two or more r.v.'s corresponds to measuring this variable for two or more objects!

Recall: From the 1st week of classes, we motivated probability by wanting to understand how likely a particular sample is (from a population).

In describing single r.v.'s at a time, we can only make statements about a sample comprised of one object.

To say any more, we must study ways of characterizing several r.v.'s at once.

$\Rightarrow$ We need to study jointly distributed random variables.

Joint Distributions: Case I: Discrete RV's

As with single r.v.'s, we describe joint discrete r.v.'s by the prob. of each outcome.

Def. (Joint PMF): Let X, Y be two discrete r.v.'s. Their joint prob. mass fn., denoted $p(x, y)$ satisfies:

$$p(x, y) = \mathbb{P}(X=x, \text{ and } Y=y), \text{ for any } x \in \text{Range}(X) \text{ and } y \in \text{Range}(Y)$$

Properties of Joint pmf:

- $0 \leq p(x, y) \leq 1$
- $\sum_{x, y} p(x, y) = 1$
- For any set A comprised of pairs of (x, y) values,

$$\mathbb{P}((X, Y) \in A) = \sum_{(x, y) \in A} p(x, y)$$

Ex: Joint pmf's for 2 rv's are often described by a grid

		y		
p(x,y)		0	1	2
x	0	$\frac{1}{100}$	0	0
	1	$\frac{9}{100}$	$\frac{9}{100}$	0
	2	0	0	$\frac{81}{100}$

← = $P(X=2, \text{ and } Y=2)$

If $A = \{(0,2), (1,1), (2,0)\}$, then:

$$\begin{aligned}
 P((X,Y) \in A) &= P((X,Y)=(0,2)) + P((X,Y)=(1,1)) + P((X,Y)=(2,0)) \quad (\text{by Axiom 3}) \\
 &= p(0,2) + p(1,1) + p(2,0) \\
 &= \underline{\underline{0.09}}
 \end{aligned}$$

Now, let $B = \{(1,0), (1,1), (1,2)\}$, i.e. B = "event that $X=1$ ". Then:

$$\begin{aligned}
 P((X,Y) \in B) &= p(1,0) + p(1,1) + p(1,2) \\
 &= \underline{\underline{0.18}} = \underline{\underline{P(X=1)}}
 \end{aligned}$$

↑ Since B contains exactly all outcomes with $X=1$!

Def. (Marginal PMF): The marginal pmf of X , denoted $p_X(\cdot)$ is equal to

$$p_X(x) = \sum_{y \in \text{Range}(Y)} p(x,y).$$

Similarly the marginal pmf of Y is $p_Y(y) = \sum_{x \in \text{Range}(X)} p(x,y)$

Ex: With joint pmf as above, the marginal pmf's are:

		y		
p(x,y)		0	1	2
x	0	$\frac{1}{100}$	0	0
	1	$\frac{9}{100}$	$\frac{9}{100}$	0
	2	0	0	$\frac{81}{100}$

 $\Rightarrow$

x	0	1	2
$p_X(x)$	$\frac{1}{100}$	$\frac{18}{100}$	$\frac{81}{100}$

$p(1,0) + p(1,1) + p(1,2)$
 $\frac{9}{100} + \frac{9}{100} + 0$

y	0	1	2
$p_Y(y)$	$\frac{10}{100}$	$\frac{9}{100}$	$\frac{81}{100}$

We would like to study how the distributions of these two (or more) r.v.'s changes when we condition on some information. For instance, what is the distribution of X given $Y=1$?

Def. (Conditional PMF): If X, Y are discrete r.v.'s with joint pmf p , then the conditional pmf of X given $Y=y$ is equal to:

$$P_{X|Y}(x|y) = \frac{p(x, y)}{P_Y(y)}, \text{ for all } x \in \text{Range}(X),$$

for all values of y for which $\underline{P_Y(y)} > 0$. ← marginal pmf of Y

Note: $p(x, y) = P(X=x, \text{ and } Y=y)$, and $P_Y(y) = P(Y=y)$.

$$\text{So } P_{X|Y}(x|y) = \frac{P(X=x, \text{ and } Y=y)}{P(Y=y)} = P(X=x | Y=y)$$

↙ by def'n of conditional probability

★ **Ex:** Consider the following joint pmf.

$p(x, y)$		y				Marginal of X :	
		0	1	2	3	x	$P_X(x)$
x	0	.08 (+)	.07 (+)	.03 (+)	.01	0	(=) 0.19
	1	<u>.07</u>	<u>.14</u>	<u>.07</u>	<u>.03</u>	1	<u>0.31</u>
	2	.03	.07	.14	.07	2	0.31
	3	.01	.03	.07	.08	3	0.19

⇒

So, $P_{Y|X}(y|1)$ is given by:

y	0	1	2	3
$P_{Y X}(y 1)$	$\frac{.07}{.31}$	$\frac{.14}{.31}$	$\frac{.07}{.31}$	$\frac{.03}{.31}$

Notice that this is different from the marginal pmf of Y !

y	0	1	2	3
$P_Y(y)$	0.19	0.31	0.31	0.19

Hence, observing/knowing that $X=1$ changed the distribution of Y .

Formally, the events $\{X=1\}$ and $\{Y=y\}$, for $y=0,1,2,3$, are not independent.

$\Rightarrow$ So we should believe that the random variables X, Y are also not independent!

Def. (Independent R.V.'s - Discrete): Discrete r.v.'s X, Y are said to be independent if $p(x, y) = p_X(x) p_Y(y)$ for any x, y .

Observation #1: So if we are told that X, Y are independent, and we know each of their marginal dist's, then we can easily compute their joint distribution!

Obs. #2: If X, Y are independent, then $p_{X|Y}(x|y) = p_X(x)$ for all x and any y !

Ex: In the above $\textcircled{\otimes}$, X and Y are not independent.

What if we have more than two discrete r.v.'s?
Can we describe a joint pmf for them?

Def. (Joint PMF, n discrete r.v.'s): The fn. $p(x_1, x_2, \dots, x_n)$ is the joint pmf of $X_1, X_2, \dots, X_n$, if:

$$p(x_1, x_2, \dots, x_n) = P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

for all $x_1, x_2, \dots, x_n$.

If we now have more than two r.v.'s, to compute the marginal pmf's, we have to add up the joint pmf over all other random variables.

Def. (Marginal PMF, n discrete r.v.'s): If $X_1, X_2, \dots, X_n$ have joint pmf p , then the marginal pmf of X_k , for $k=1, 2, \dots, n$,

is:

$$p_{X_k}(x_k) = P(X_k = x_k) = \sum_{x_1} \sum_{x_2} \dots \sum_{x_{k-1}} \sum_{x_{k+1}} \dots \sum_{x_n} p(x_1, x_2, \dots, x_n)$$

Note: the k^{th} argument here is always the same in every term.

Note: Each term in the sum represents some different outcome of all $X_1, \dots, X_n$ in which $X_k = x_k$. We want to add up the probabilities of all such outcomes.

Now we can turn to the more general notion of independence for discrete r.v.'s.

Def. (Mutually Indep. R.V.'s - Discrete): Let $X_1, \dots, X_n$ be discrete r.v.'s with joint pmf $p(x_1, x_2, \dots, x_n)$. We say that they are mutually independent if $p(x_1, x_2, \dots, x_n) = p_{X_1}(x_1) \cdot p_{X_2}(x_2) \cdot \dots \cdot p_{X_n}(x_n)$ for all $x_1, \dots, x_n$. ← marginal pmf of X_n

Note: This means that the n events $\{X_1 = x_1\}, \{X_2 = x_2\}, \dots, \{X_n = x_n\}$ will be mutually independent for any choice of $x_1, \dots, x_n$!

Case I: Continuous RV's

With 2 continuous r.v.'s, X and Y , we will still have

$$\mathbb{P}(X=x, Y=y) = 0 \text{ for any } x, y.$$

So as before, we must describe the joint distribution of continuous r.v.'s through a density.

But instead of obtaining the prob. of a (1-dimensional) interval, we use the density to compute the prob. of (2-dimensional) rectangles.

Def. (Joint PDF): If X, Y are two continuous r.v.'s, their joint pdf, $f(x, y)$, must satisfy:

- $f(x, y) \geq 0$, for all x, y

- $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$

- For any two intervals $[a_1, b_1], [a_2, b_2]$:

$$\mathbb{P}(X \in [a_1, b_1], Y \in [a_2, b_2]) = \int_{a_1}^{b_1} \int_{a_2}^{b_2} f(x, y) dy dx$$